

Lab: Neural Bases of Social Communication

Learning Goal: Map social cognition to neural circuits and analyze communication errors at the neural level.

Part 1: Social Brain Network Mapping

Exercise: For each social task, identify the primary brain regions involved and explain their role:

Social Task	Primary Regions	What is each doing?
Recognizing a friend's face		
Inferring someone is lying		
Understanding a joke		
Feeling empathy for a crying child		
Following someone's gaze direction		

Write your response here

Part 2: Language Prediction Errors

Learning Goal: Identify ERPs as prediction errors in language processing.

Exercise: For each sentence, predict whether you'd see an N400 (semantic surprise), P600 (syntactic surprise), both, or neither:

- "The cat sat on the mat." →

2. "The cat sat on the *algebra*." →
3. "The cats *was* sitting on the mat." →
4. "The cat sat on the *dogmatically*." →
5. "She spread the warm bread with *socks*." →

Explain your reasoning for each.

Write your response here

Part 3: Mirror vs. Mentalizing Systems

Learning Goal: Distinguish the two routes to social understanding.

Scenarios: For each, indicate whether the mirror system, mentalizing network, or both are primarily engaged, and why:

1. Watching a dancer perform and feeling the movements in your own body
2. Figuring out why your friend seems upset despite saying "I'm fine"
3. Yawning after seeing someone else yawn
4. Understanding a character's motivation in a novel

Write your response here

Part 4: Oxytocin and Social Precision

Learning Goal: Analyze how oxytocin modulates social inference.

Research analysis: Studies show that intranasal oxytocin increases:

- Eye contact during conversation
- Trust in economic games
- Ability to detect subtle emotions in facial expressions

For each finding, explain: How does increasing social precision (gain on social signals) produce this result?

Write your response here

Part 5: Reflection

In 150 words, reflect: If autism involves different precision weighting rather than "broken" social cognition, how should this change how we think about autism? What are the strengths of a brain that weights sensory detail more heavily than social priors?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Region-function mapping	Social brain network
2	ERP prediction	Language processing as predictive coding
3	System distinction	Mirror vs. mentalizing routes
4	Research interpretation	Oxytocin and social precision
5	Ethical reflection	Neurodiversity and precision profiles